

Conducting Forensic Investigations on Linux Systems (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 06

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Time on Task:

3 hours, 57 minutes

Progress:

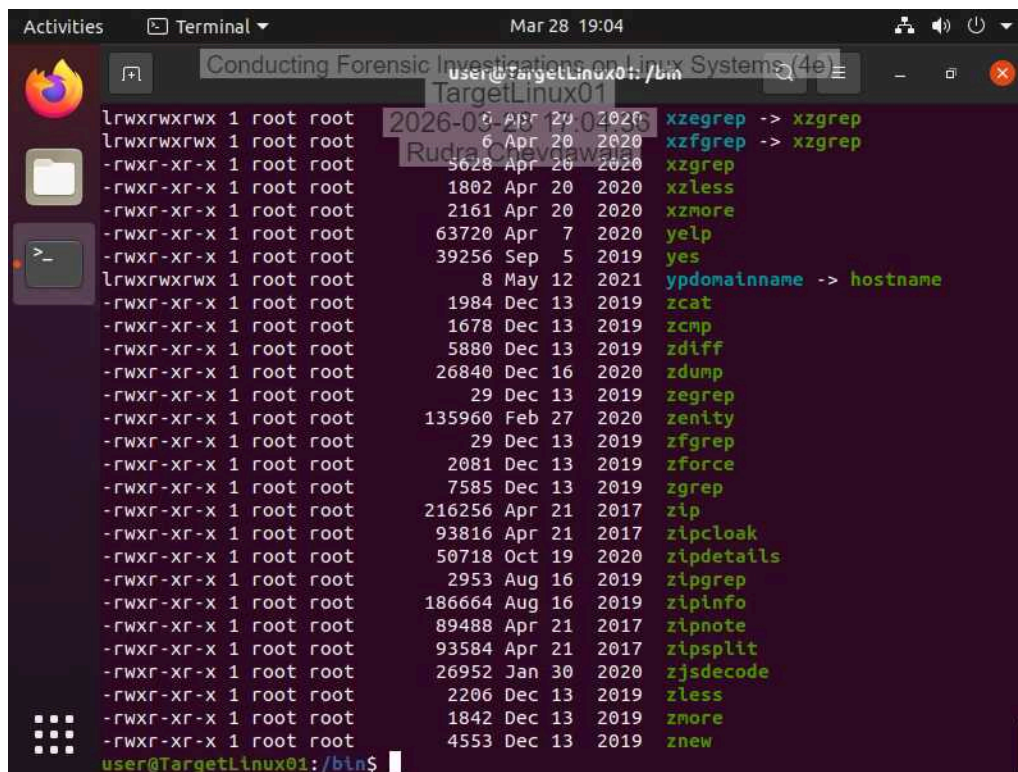
100%

Report Generated: Sunday, March 29, 2026 at 1:39 PM

Section 1: Hands-On Demonstration

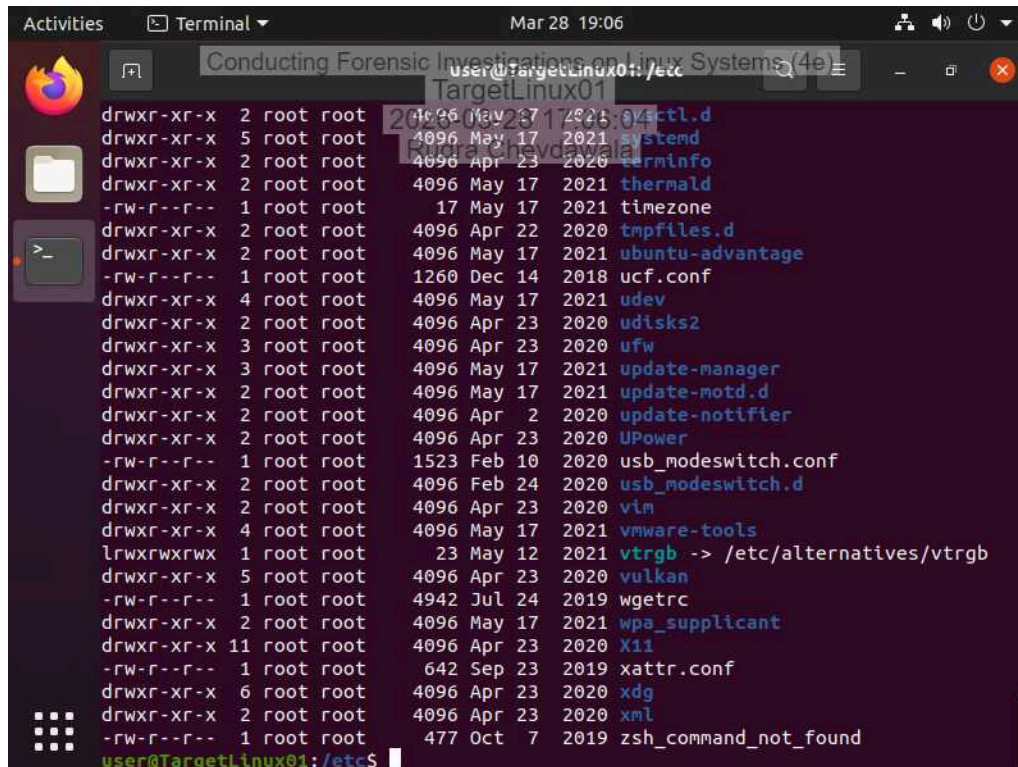
Part 1: Explore a Live Linux System

17. Make a screen capture showing the contents of the `/bin` directory.



```
Activities Terminal Mar 28 19:04
user@TargetLinux01: /bin$ ls -l
lrwxrwxrwx 1 root root 5 Apr 20 2020 xzgrep -> xzgrep
lrwxrwxrwx 1 root root 6 Apr 20 2020 xzfgrep -> xzfgrep
-rwxr-xr-x 1 root root 5628 Apr 20 2020 xzgrep
-rwxr-xr-x 1 root root 1802 Apr 20 2020 xzless
-rwxr-xr-x 1 root root 2161 Apr 20 2020 xzmore
-rwxr-xr-x 1 root root 63720 Apr 7 2020 yelp
-rwxr-xr-x 1 root root 39256 Sep 5 2019 yes
lrwxrwxrwx 1 root root 8 May 12 2021 ypdomainname -> hostname
-rwxr-xr-x 1 root root 1984 Dec 13 2019 zcat
-rwxr-xr-x 1 root root 1678 Dec 13 2019 zcmp
-rwxr-xr-x 1 root root 5880 Dec 13 2019 zdiff
-rwxr-xr-x 1 root root 26840 Dec 16 2020 zdump
-rwxr-xr-x 1 root root 29 Dec 13 2019 zegrep
-rwxr-xr-x 1 root root 135960 Feb 27 2020 zenity
-rwxr-xr-x 1 root root 29 Dec 13 2019 zfgrep
-rwxr-xr-x 1 root root 2081 Dec 13 2019 zforce
-rwxr-xr-x 1 root root 7585 Dec 13 2019 zgrep
-rwxr-xr-x 1 root root 216256 Apr 21 2017 zip
-rwxr-xr-x 1 root root 93816 Apr 21 2017 zipcloak
-rwxr-xr-x 1 root root 50718 Oct 19 2020 zipdetails
-rwxr-xr-x 1 root root 2953 Aug 16 2019 zipgrep
-rwxr-xr-x 1 root root 186664 Aug 16 2019 zipinfo
-rwxr-xr-x 1 root root 89488 Apr 21 2017 zipnote
-rwxr-xr-x 1 root root 93584 Apr 21 2017 zipsplit
-rwxr-xr-x 1 root root 26952 Jan 30 2020 zjsdecode
-rwxr-xr-x 1 root root 2206 Dec 13 2019 zless
-rwxr-xr-x 1 root root 1842 Dec 13 2019 zmore
-rwxr-xr-x 1 root root 4553 Dec 13 2019 znew
user@TargetLinux01: /bin$
```

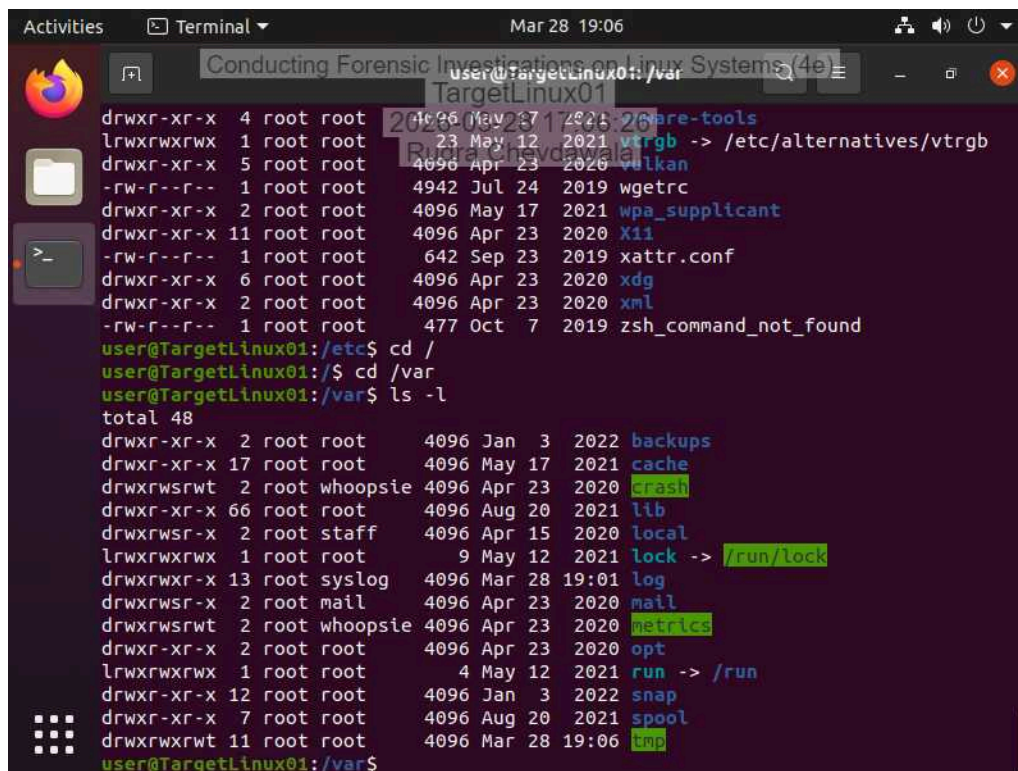
20. Make a screen capture showing the contents of the /etc directory.



A terminal window titled "Terminal" with a search bar containing "Conducting Forensic Investigations on Linux Systems (4e)". The terminal shows the command `ls -l /etc` and its output, listing files in the /etc directory. The output is as follows:

```
drwxr-xr-x 2 root root 4096 May 27 2021 etc.d
drwxr-xr-x 5 root root 4096 May 17 2021 systemd
drwxr-xr-x 2 root root 4096 Apr 23 2020 terminfo
drwxr-xr-x 2 root root 4096 May 17 2021 thermald
-rw-r--r-- 1 root root 17 May 17 2021 timezone
drwxr-xr-x 2 root root 4096 Apr 22 2020 tmpfiles.d
drwxr-xr-x 2 root root 4096 May 17 2021 ubuntu-advantage
-rw-r--r-- 1 root root 1260 Dec 14 2018 ucf.conf
drwxr-xr-x 4 root root 4096 May 17 2021 udev
drwxr-xr-x 2 root root 4096 Apr 23 2020 udisks2
drwxr-xr-x 3 root root 4096 Apr 23 2020 ufw
drwxr-xr-x 3 root root 4096 May 17 2021 update-manager
drwxr-xr-x 2 root root 4096 May 17 2021 update-motd.d
drwxr-xr-x 2 root root 4096 Apr 2 2020 update-notifier
drwxr-xr-x 2 root root 4096 Apr 23 2020 UPower
-rw-r--r-- 1 root root 1523 Feb 10 2020 usb_modeswitch.conf
drwxr-xr-x 2 root root 4096 Feb 24 2020 usb_modeswitch.d
drwxr-xr-x 2 root root 4096 Apr 23 2020 vim
drwxr-xr-x 4 root root 4096 May 17 2021 vmware-tools
lrwxrwxrwx 1 root root 23 May 12 2021 vtrgb -> /etc/alternatives/vtrgb
drwxr-xr-x 5 root root 4096 Apr 23 2020 vulkan
-rw-r--r-- 1 root root 4942 Jul 24 2019 wgetrc
drwxr-xr-x 2 root root 4096 May 17 2021 wpa_supplicant
drwxr-xr-x 11 root root 4096 Apr 23 2020 X11
-rw-r--r-- 1 root root 642 Sep 23 2019 xattr.conf
drwxr-xr-x 6 root root 4096 Apr 23 2020 xdg
drwxr-xr-x 2 root root 4096 Apr 23 2020 xml
-rw-r--r-- 1 root root 477 Oct 7 2019 zsh_command_not_found
```

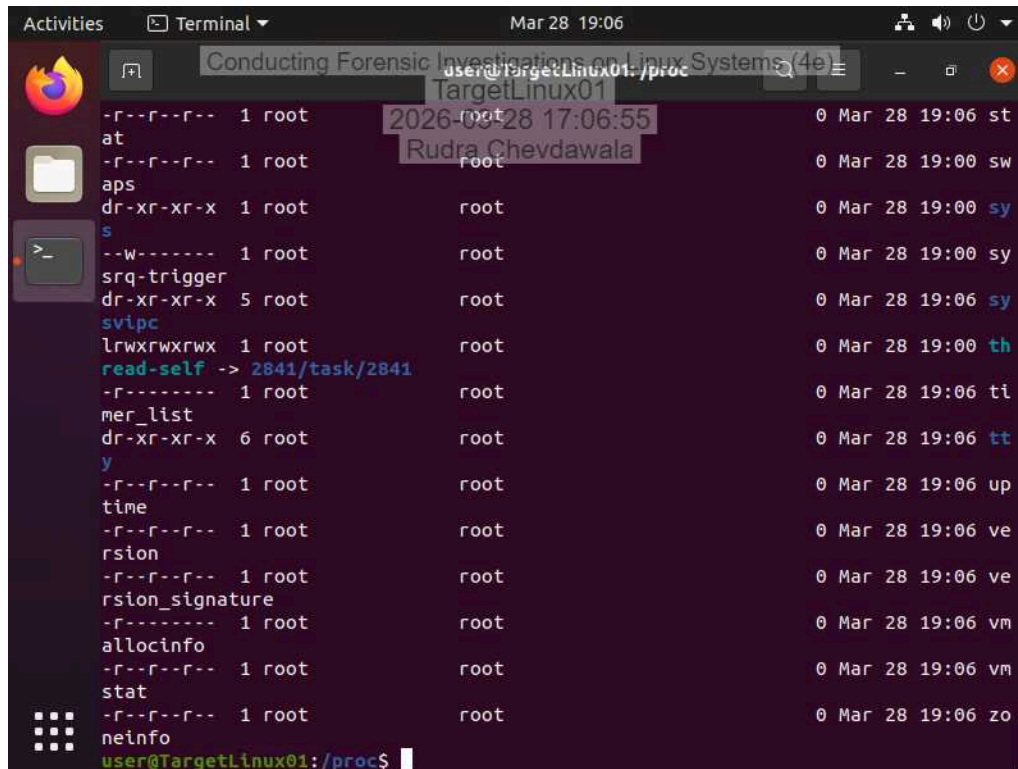
21. Make a screen capture showing the contents of the /var directory.



A terminal window titled "Terminal" with a search bar containing "Conducting Forensic Investigations on Linux Systems (4e)". The terminal shows the command `ls -l /var` and its output, listing files in the /var directory. The output is as follows:

```
total 48
drwxr-xr-x 2 root root 4096 Jan 3 2022 backups
drwxr-xr-x 17 root root 4096 May 17 2021 cache
drwxrwsrwt 2 root whoopsie 4096 Apr 23 2020 crash
drwxr-xr-x 66 root root 4096 Aug 20 2021 lib
drwxrwsr-x 2 root staff 4096 Apr 15 2020 local
lrwxrwxrwx 1 root root 9 May 12 2021 lock -> /run/lock
drwxrwsr-x 13 root syslog 4096 Mar 28 19:01 log
drwxrwsr-x 2 root mail 4096 Apr 23 2020 mail
drwxrwsrwt 2 root whoopsie 4096 Apr 23 2020 metrics
drwxr-xr-x 2 root root 4096 Apr 23 2020 opt
lrwxrwxrwx 1 root root 4 May 12 2021 run -> /run
drwxr-xr-x 12 root root 4096 Jan 3 2022 snap
drwxr-xr-x 7 root root 4096 Aug 20 2021 spool
drwxrwsrwt 11 root root 4096 Mar 28 19:06 tmp
```

22. Make a screen capture showing the contents of the `/proc` directory.

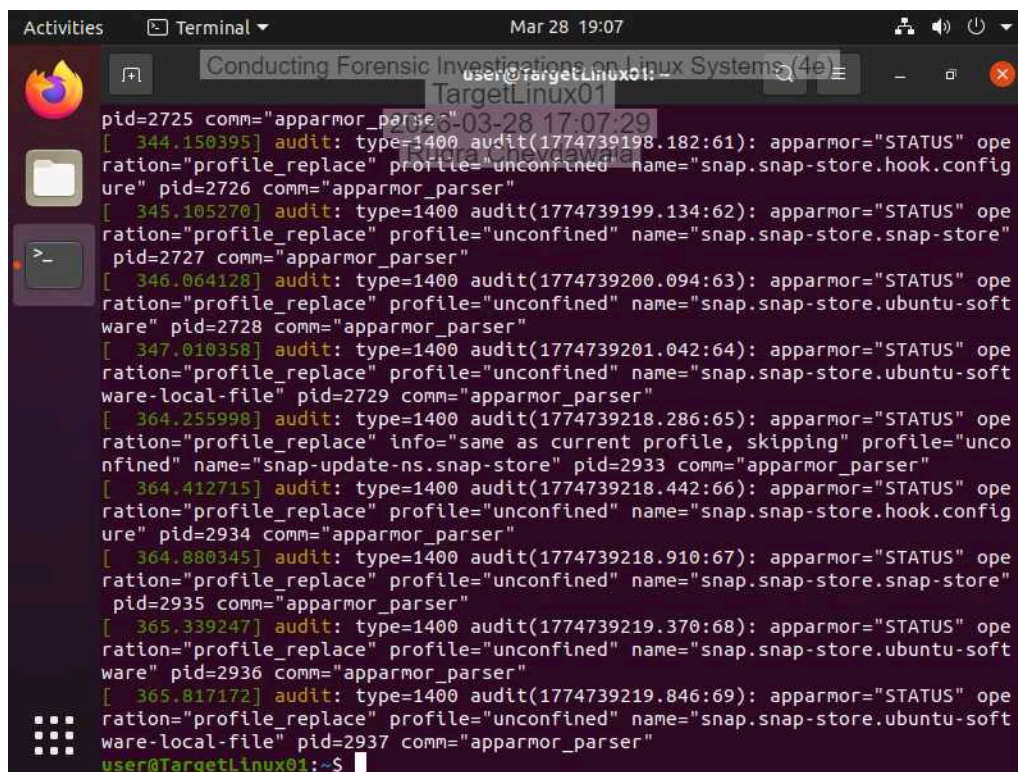


A terminal window titled "Terminal" showing the contents of the `/proc` directory. The window has a dark background with light-colored text. The title bar includes "Activities", "Terminal", and the date/time "Mar 28 19:06". The terminal output lists various system files and their permissions, owner, and size. The files listed are: `at`, `aps`, `dr-xr-xr-x`, `svipc`, `lrwxrwxrwx`, `read-self -> 2841/task/2841`, `mer_list`, `dr-xr-xr-x`, `y`, `time`, `rsion`, `rsion_signature`, `allocinfo`, `stat`, and `neinfo`. The permissions are shown in columns, followed by the owner (root), the size (1 or 5), and the file name. The terminal prompt is `user@TargetLinux01:/proc$`.

```
user@TargetLinux01:/proc$ ls -la
-r--r--r-- 1 root 0 Mar 28 19:06 st
at
-r--r--r-- 1 root 0 Mar 28 19:00 sw
aps
dr-xr-xr-x 1 root 0 Mar 28 19:00 sy
svipc
lrwxrwxrwx 1 root 0 Mar 28 19:00 sy
read-self -> 2841/task/2841
-r--r--r-- 1 root 0 Mar 28 19:06 ti
mer_list
dr-xr-xr-x 6 root 0 Mar 28 19:06 tt
y
-r--r--r-- 1 root 0 Mar 28 19:06 up
time
-r--r--r-- 1 root 0 Mar 28 19:06 ve
rsion
-r--r--r-- 1 root 0 Mar 28 19:06 ve
rsion_signature
-r--r--r-- 1 root 0 Mar 28 19:06 vm
allocinfo
-r--r--r-- 1 root 0 Mar 28 19:06 vm
stat
-r--r--r-- 1 root 0 Mar 28 19:06 zo
neinfo
user@TargetLinux01:/proc$
```

Part 2: Use Linux Shell Commands for Forensic Investigations

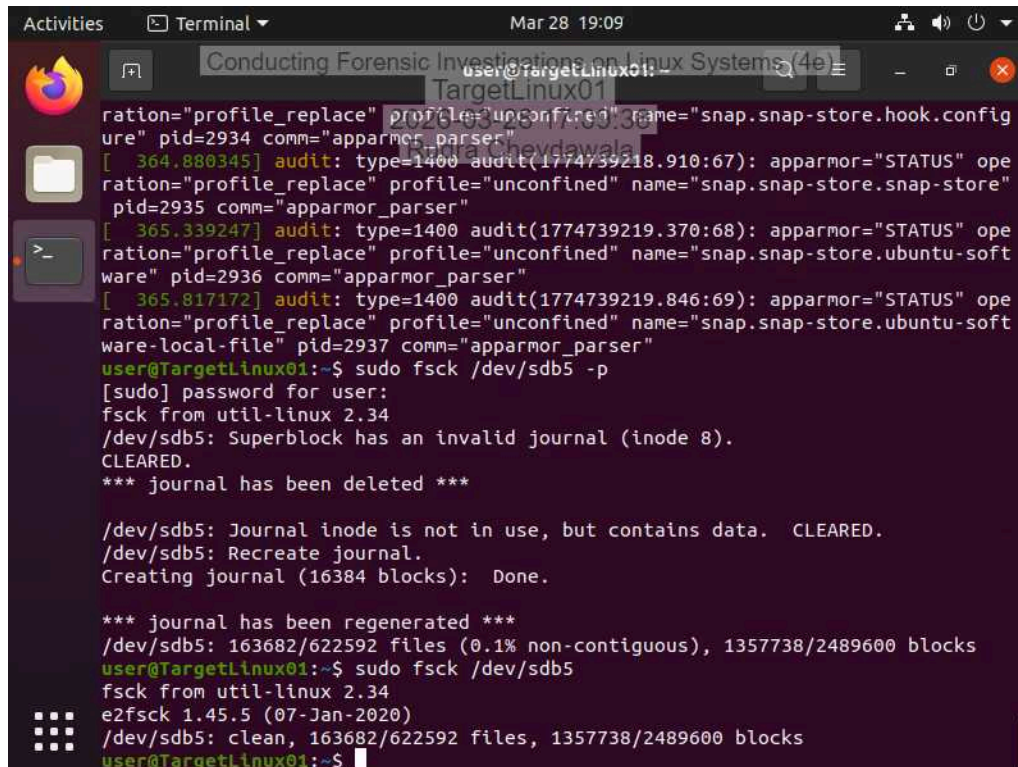
2. Make a screen capture showing the results of the `dmesg` command.



The screenshot shows a terminal window titled "Terminal" with the date and time "Mar 28 19:07". The terminal displays the output of the `dmesg` command, showing a series of audit messages from the `apparmor_parser` process. The messages are timestamped and include details about profile replacements and operations. The terminal prompt is `user@TargetLinux01:~$`.

```
pid=2725 comm="apparmor_parser"
[ 344.150395] audit: type=1400 audit(1774739198.182:61): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.hook.config" pid=2726 comm="apparmor_parser"
[ 345.105270] audit: type=1400 audit(1774739199.134:62): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.snap-store" pid=2727 comm="apparmor_parser"
[ 346.064128] audit: type=1400 audit(1774739200.094:63): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software" pid=2728 comm="apparmor_parser"
[ 347.010358] audit: type=1400 audit(1774739201.042:64): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software-local-file" pid=2729 comm="apparmor_parser"
[ 364.255998] audit: type=1400 audit(1774739218.286:65): apparmor="STATUS" operation="profile_replace" info="same as current profile, skipping" profile="unconfined" name="snap-update-ns.snap-store" pid=2933 comm="apparmor_parser"
[ 364.412715] audit: type=1400 audit(1774739218.442:66): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.hook.config" pid=2934 comm="apparmor_parser"
[ 364.880345] audit: type=1400 audit(1774739218.910:67): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.snap-store" pid=2935 comm="apparmor_parser"
[ 365.339247] audit: type=1400 audit(1774739219.370:68): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software" pid=2936 comm="apparmor_parser"
[ 365.817172] audit: type=1400 audit(1774739219.846:69): apparmor="STATUS" operation="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-software-local-file" pid=2937 comm="apparmor_parser"
user@TargetLinux01:~$
```

7. Make a screen capture showing the results of the fsck command.

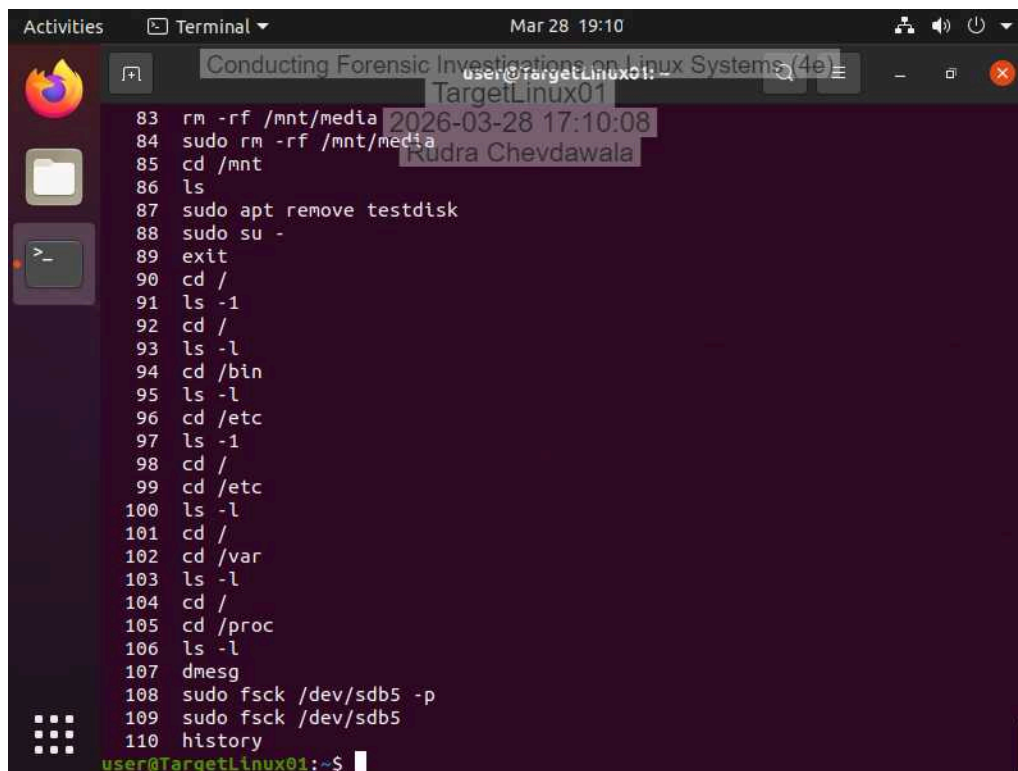


```
ration="profile_replace" profile="unconfined" name="snap.snap-store.hook.config
ure" pid=2934 comm="apparmor_parser"
[ 364.880345] audit: type=1400 audit(1774739218.910:67): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap.snap-store.snap-store"
pid=2935 comm="apparmor_parser"
[ 365.339247] audit: type=1400 audit(1774739219.370:68): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-soft
ware" pid=2936 comm="apparmor_parser"
[ 365.817172] audit: type=1400 audit(1774739219.846:69): apparmor="STATUS" ope
ration="profile_replace" profile="unconfined" name="snap.snap-store.ubuntu-soft
ware-local-file" pid=2937 comm="apparmor_parser"
user@TargetLinux01:~$ sudo fsck /dev/sdb5 -p
[sudo] password for user:
fsck from util-linux 2.34
/dev/sdb5: Superblock has an invalid journal (inode 8).
CLEARED.
*** journal has been deleted ***

/dev/sdb5: Journal inode is not in use, but contains data.  CLEARED.
/dev/sdb5: Recreate journal.
Creating journal (16384 blocks):  Done.

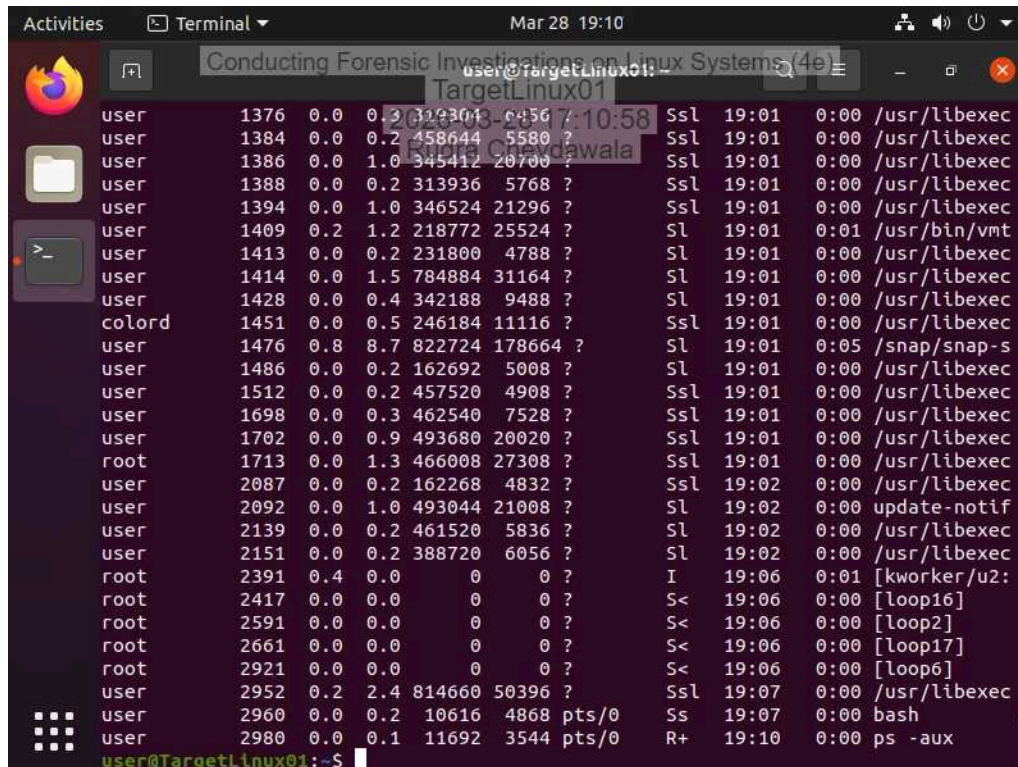
*** journal has been regenerated ***
/dev/sdb5: 163682/622592 files (0.1% non-contiguous), 1357738/2489600 blocks
user@TargetLinux01:~$ sudo fsck /dev/sdb5
fsck from util-linux 2.34
e2fsck 1.45.5 (07-Jan-2020)
/dev/sdb5: clean, 163682/622592 files, 1357738/2489600 blocks
user@TargetLinux01:~$
```

9. Make a screen capture showing the results of the history command.



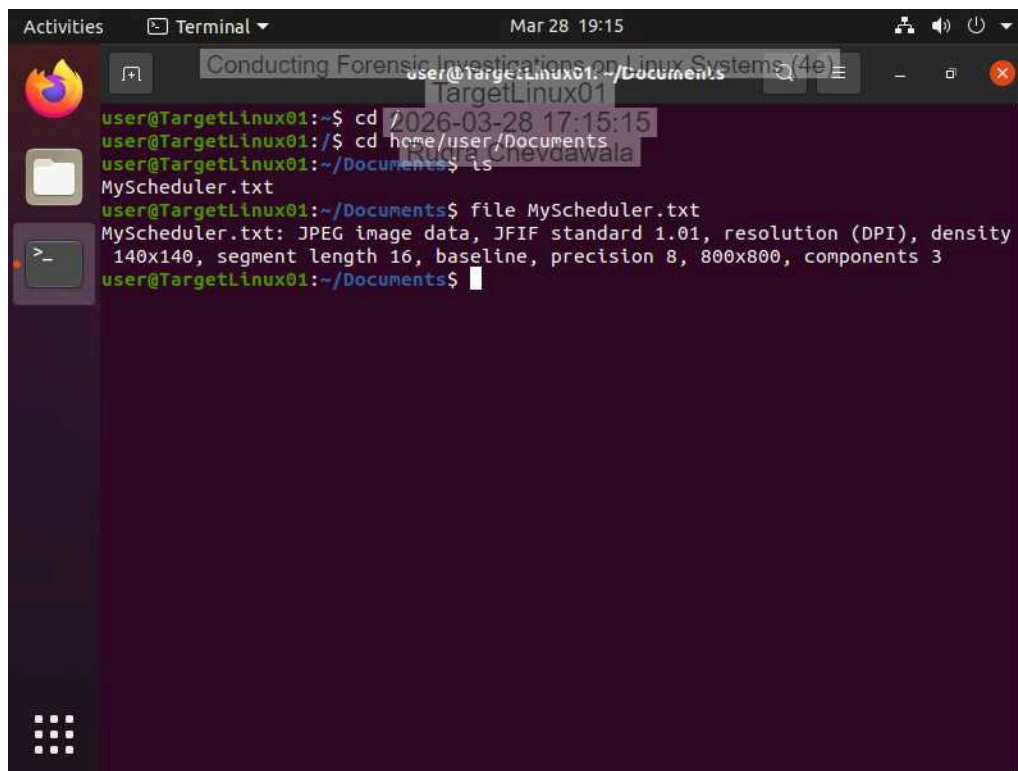
```
83 rm -rf /mnt/media
84 sudo rm -rf /mnt/media
85 cd /mnt
86 ls
87 sudo apt remove testdisk
88 sudo su -
89 exit
90 cd /
91 ls -l
92 cd /
93 ls -l
94 cd /bin
95 ls -l
96 cd /etc
97 ls -l
98 cd /
99 cd /etc
100 ls -l
101 cd /
102 cd /var
103 ls -l
104 cd /
105 cd /proc
106 ls -l
107 dmesg
108 sudo fsck /dev/sdb5 -p
109 sudo fsck /dev/sdb5
110 history
user@TargetLinux01:~$
```

11. Make a screen capture showing the running processes.



```
user@TargetLinux01:~$ ps -aux
user      1376  0.0  0.3 312304  6456 ?        Ssl  19:01   0:00 /usr/libexec
user      1384  0.0  0.2 458644  5580 ?        Ssl  19:01   0:00 /usr/libexec
user      1386  0.0  1.0 345412 20700 ?        Ssl  19:01   0:00 /usr/libexec
user      1388  0.0  0.2 313936  5768 ?        Ssl  19:01   0:00 /usr/libexec
user      1394  0.0  1.0 346524 21296 ?        Ssl  19:01   0:00 /usr/libexec
user      1409  0.2  1.2 218772 25524 ?        Sl   19:01   0:01 /usr/bin/vmt
user      1413  0.0  0.2 231800  4788 ?        Sl   19:01   0:00 /usr/libexec
user      1414  0.0  1.5 784884 31164 ?        Sl   19:01   0:00 /usr/libexec
user      1428  0.0  0.4 342188  9488 ?        Sl   19:01   0:00 /usr/libexec
colord    1451  0.0  0.5 246184 11116 ?        Ssl  19:01   0:00 /usr/libexec
user      1476  0.8  8.7 822724 178664 ?        Sl   19:01   0:05 /snap/snap-s
user      1486  0.0  0.2 162692  5008 ?        Sl   19:01   0:00 /usr/libexec
user      1512  0.0  0.2 457520  4908 ?        Ssl  19:01   0:00 /usr/libexec
user      1698  0.0  0.3 462540  7528 ?        Ssl  19:01   0:00 /usr/libexec
user      1702  0.0  0.9 493680 20020 ?        Ssl  19:01   0:00 /usr/libexec
root      1713  0.0  1.3 466008 27308 ?        Ssl  19:01   0:00 /usr/libexec
user      2087  0.0  0.2 162268  4832 ?        Ssl  19:02   0:00 /usr/libexec
user      2092  0.0  1.0 493044 21008 ?        Sl   19:02   0:00 update-notif
user      2139  0.0  0.2 461520  5836 ?        Sl   19:02   0:00 /usr/libexec
user      2151  0.0  0.2 388720  6056 ?        Sl   19:02   0:00 /usr/libexec
root      2391  0.4  0.0  0 0 ?        I    19:06   0:01 [kworker/u2:
root      2417  0.0  0.0  0 0 ?        S<   19:06   0:00 [loop16]
root      2591  0.0  0.0  0 0 ?        S<   19:06   0:00 [loop2]
root      2661  0.0  0.0  0 0 ?        S<   19:06   0:00 [loop17]
root      2921  0.0  0.0  0 0 ?        S<   19:06   0:00 [loop6]
user      2952  0.2  2.4 814660 50396 ?        Ssl  19:07   0:00 /usr/libexec
user      2960  0.0  0.2 10616  4868 pts/0    Ss   19:07   0:00 bash
user      2980  0.0  0.1 11692  3544 pts/0    R+   19:10   0:00 ps -aux
```

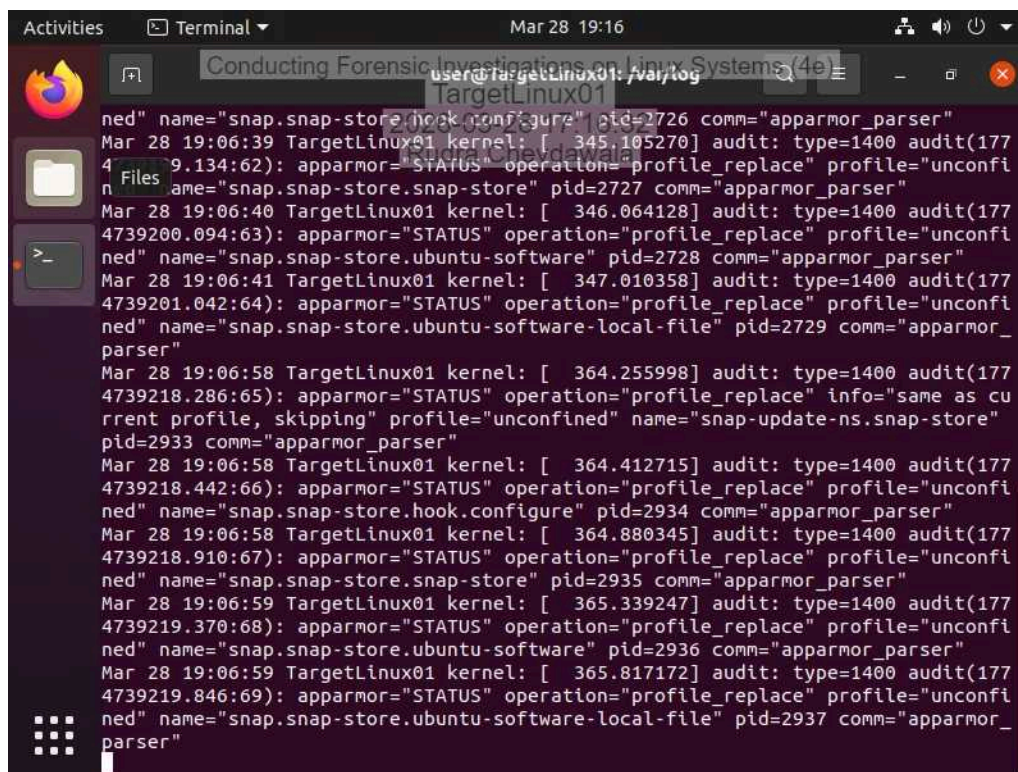
15. Make a screen capture showing the results of the file command.



```
user@TargetLinux01:~$ cd /home/user/Documents
user@TargetLinux01:~/Documents$ ls
MyScheduler.txt
user@TargetLinux01:~/Documents$ file MyScheduler.txt
MyScheduler.txt: JPEG image data, JFIF standard 1.01, resolution (DPI), density
140x140, segment length 16, baseline, precision 8, 800x800, components 3
```


Part 3: Retrieve Logs Files on a Live Linux System

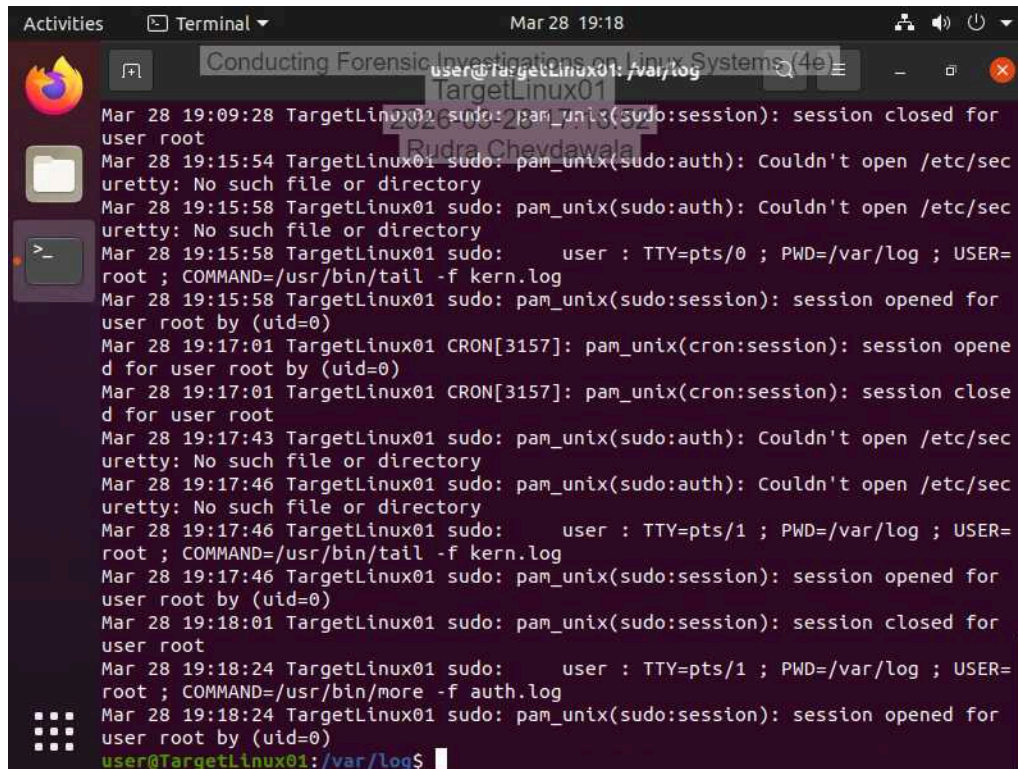
4. Make a screen capture showing the records in the kern.log file.



The screenshot shows a terminal window titled "Terminal" with the date and time "Mar 28 19:16". The terminal displays the contents of the `kern.log` file, showing a series of audit records. The records are timestamped "Mar 28 19:06:39" through "Mar 28 19:06:59" and originate from "TargetLinux01 kernel". Each record includes an audit type (1400), an audit ID (1774), and details about an AppArmor profile replacement operation. The records show the status of the operation, the profile name ("unconfined"), and the name of the snap component being updated. The terminal window has a dark background with light-colored text. On the left side of the terminal, there is a sidebar with icons for "Activities", "Files", and a terminal icon. The top of the terminal window shows the title bar with "Terminal" and the date/time. The bottom of the terminal window shows a prompt character and a cursor.

```
ned" name="snap.snap-store.hook.configure" pid=2726 comm="apparmor_parser"
Mar 28 19:06:39 TargetLinux01 kernel: [ 345.105270] audit: type=1400 audit(1774
9.134:62): apparmor="STATUS" operation="profile_replace" profile="unconfi
n" name="snap.snap-store.snap-store" pid=2727 comm="apparmor_parser"
Mar 28 19:06:40 TargetLinux01 kernel: [ 346.064128] audit: type=1400 audit(1774
739200.094:63): apparmor="STATUS" operation="profile_replace" profile="unconfi
ned" name="snap.snap-store.ubuntu-software" pid=2728 comm="apparmor_parser"
Mar 28 19:06:41 TargetLinux01 kernel: [ 347.010358] audit: type=1400 audit(1774
739201.042:64): apparmor="STATUS" operation="profile_replace" profile="unconfi
ned" name="snap.snap-store.ubuntu-software-local-file" pid=2729 comm="apparmor_
parser"
Mar 28 19:06:58 TargetLinux01 kernel: [ 364.255998] audit: type=1400 audit(1774
739218.286:65): apparmor="STATUS" operation="profile_replace" info="same as cu
rrent profile, skipping" profile="unconfined" name="snap-update-ns.snap-store"
pid=2933 comm="apparmor_parser"
Mar 28 19:06:58 TargetLinux01 kernel: [ 364.412715] audit: type=1400 audit(1774
739218.442:66): apparmor="STATUS" operation="profile_replace" profile="unconfi
ned" name="snap.snap-store.hook.configure" pid=2934 comm="apparmor_parser"
Mar 28 19:06:58 TargetLinux01 kernel: [ 364.880345] audit: type=1400 audit(1774
739218.910:67): apparmor="STATUS" operation="profile_replace" profile="unconfi
ned" name="snap.snap-store.snap-store" pid=2935 comm="apparmor_parser"
Mar 28 19:06:59 TargetLinux01 kernel: [ 365.339247] audit: type=1400 audit(1774
739219.370:68): apparmor="STATUS" operation="profile_replace" profile="unconfi
ned" name="snap.snap-store.ubuntu-software" pid=2936 comm="apparmor_parser"
Mar 28 19:06:59 TargetLinux01 kernel: [ 365.817172] audit: type=1400 audit(1774
739219.846:69): apparmor="STATUS" operation="profile_replace" profile="unconfi
ned" name="snap.snap-store.ubuntu-software-local-file" pid=2937 comm="apparmor_
parser"
```

7. Make a screen capture showing the records in the auth.log file.

A terminal window titled 'Terminal' with a timestamp of 'Mar 28 19:18'. The window shows a series of log entries from the 'auth.log' file. The entries are timestamped and include details about user sessions, including successful logins, session openings, and session closures. The terminal prompt is 'user@TargetLinux01:/var/log\$'.

```
Mar 28 19:09:28 TargetLinux01 sudo: pam_unix(sudo:session): session closed for user root
Mar 28 19:15:54 TargetLinux01 sudo: pam_unix(sudo:auth): Couldn't open /etc/security: No such file or directory
Mar 28 19:15:58 TargetLinux01 sudo: pam_unix(sudo:auth): Couldn't open /etc/security: No such file or directory
Mar 28 19:15:58 TargetLinux01 sudo: user : TTY=pts/0 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/tail -f kern.log
Mar 28 19:15:58 TargetLinux01 sudo: pam_unix(sudo:session): session opened for user root by (uid=0)
Mar 28 19:17:01 TargetLinux01 CRON[3157]: pam_unix(cron:session): session opened for user root by (uid=0)
Mar 28 19:17:01 TargetLinux01 CRON[3157]: pam_unix(cron:session): session closed for user root
Mar 28 19:17:43 TargetLinux01 sudo: pam_unix(sudo:auth): Couldn't open /etc/security: No such file or directory
Mar 28 19:17:46 TargetLinux01 sudo: pam_unix(sudo:auth): Couldn't open /etc/security: No such file or directory
Mar 28 19:17:46 TargetLinux01 sudo: user : TTY=pts/1 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/tail -f kern.log
Mar 28 19:17:46 TargetLinux01 sudo: pam_unix(sudo:session): session opened for user root by (uid=0)
Mar 28 19:18:01 TargetLinux01 sudo: pam_unix(sudo:session): session closed for user root
Mar 28 19:18:24 TargetLinux01 sudo: user : TTY=pts/1 ; PWD=/var/log ; USER=root ; COMMAND=/usr/bin/more -f auth.log
Mar 28 19:18:24 TargetLinux01 sudo: pam_unix(sudo:session): session opened for user root by (uid=0)
user@TargetLinux01:/var/log$
```


Section 2: Applied Learning

Part 1: Identify Login Attempts on a Linux Drive Image

15. **Document** the names of the two non-root users that attempted to log in, the number of attempts detected, the date/time range of the attempts, the source IP address for the login attempts, and the port.

The names of the two non-root users were dominic and noel. They attempted to log in 22 times from the source IP 192.168.78.1 on port 3521 between 00:57:19 and 05:39:01 on June 11th.

17. **Document** the date and time the most recent successful login for the user(s) that you previously identified in step 15.

The most recent login by dominic was on June 11th at 05:23:03.
As for noel, he was unable to log in to the system successfully, as far as I can tell.

Part 2: Identify Software Installations on a Linux Drive Image

3. **Document** the applications that were installed using apt-get, then use the Internet to identify the ones that might be considered suspicious.

Apps installed using apt-get include: logkeys, autoconf, autotools-dev, build-essentials, kdb, cacti, and openssh-server.

Applications that might be considered suspicious include logkeys, which is a keylogger capable of capturing sensitive user input. openssh-server may also be concerning, as it allows remote access to the system and could be used as a backdoor if improperly secured. Additionally, cacti, a network monitoring tool, has had known vulnerabilities that could allow remote code execution (RCE) if not patched.

Part 3: Identify External Drive Attachments on a Linux Drive Image

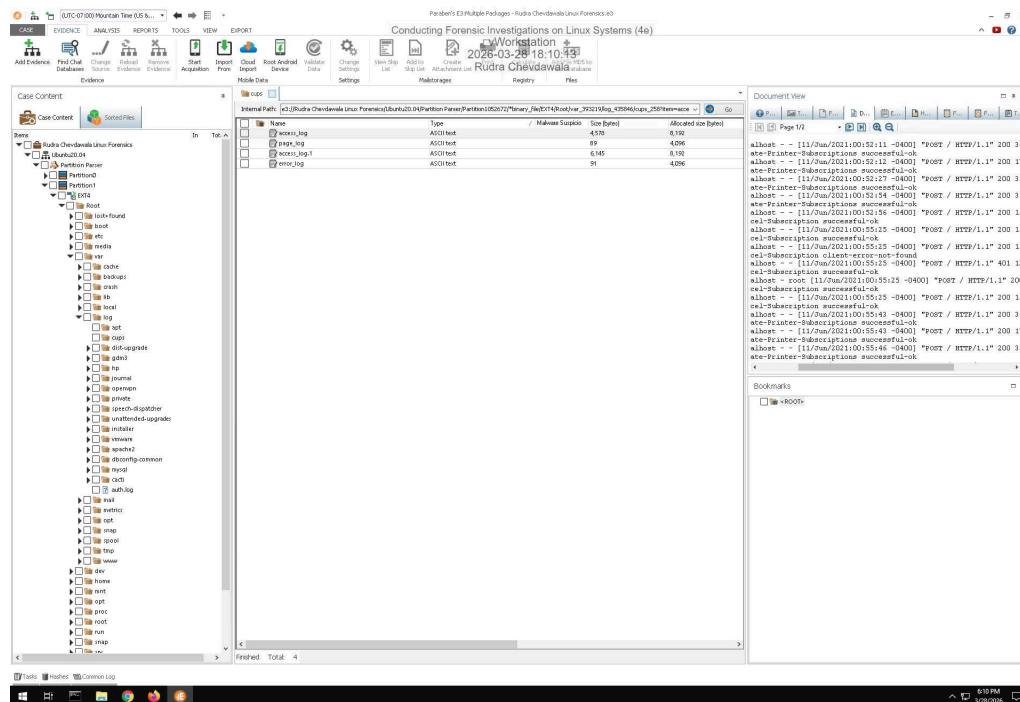
4. **Document** when the USB storage device was connected and its serial number.

The USB storage device was connected on June 10th at 10:24:17, and the serial number is FBI1405291710344.

Section 3: Challenge and Analysis

Part 1: Identify Recently Printed Files on a Linux Drive Image

Make a screen capture showing the contents of the printer log file.



Part 2: Identify Disk Imaging on a Linux Drive Image

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Make a screen capture showing the record of the dd command in the Text View.

